

Week 9 — Taylor's Theorem and Local Approximation

Motivation: Local Linear Approximation

Let

$$X \subseteq \mathbb{R}, \quad f : X \rightarrow \mathbb{R},$$

and suppose that f is differentiable.

Fix a point

$$x^* \in X.$$

Suppose we want to understand the behavior of $f(x)$ for values of x close to x^* . If f is complicated, it may be useful to approximate it locally by a simpler function.

The simplest useful approximation is a linear function

$$L(x) = a_0 + a_1x.$$

We want L to be a good approximation to f near x^* .

Intuition. Close to x^* , a differentiable function should look approximately like its tangent line. The objective is therefore to choose a linear function that agrees with f both in its value and in its slope at x^* .

The Best Linear Approximation

We impose two natural properties.

Property 1: Match the value. We require

$$L(x^*) = f(x^*).$$

Since

$$L(x) = a_0 + a_1x,$$

this means

$$a_0 + a_1x^* = f(x^*).$$

Property 2: Match the slope. We also require

$$L'(x^*) = f'(x^*).$$

Since

$$L'(x) = a_1,$$

we obtain

$$a_1 = f'(x^*).$$

Therefore

$$a_0 = f(x^*) - f'(x^*)x^*.$$

Substituting back,

$$\boxed{L(x \mid x^*) = f(x^*) + f'(x^*)(x - x^*)}.$$

This is the tangent-line approximation to f around x^* .

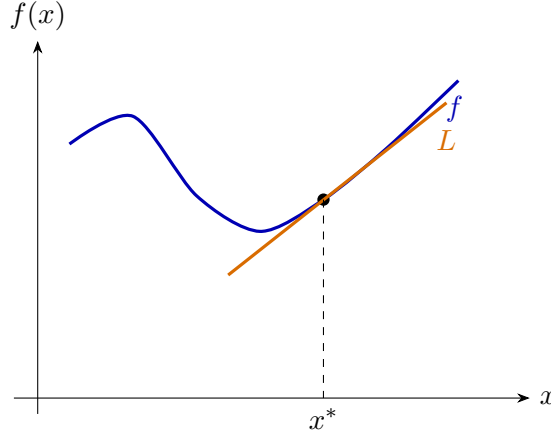


Figure 1: Near x^* , the tangent line $L(x \mid x^*) = f(x^*) + f'(x^*)(x - x^*)$ provides the first-order approximation to f .

The Remainder of the Linear Approximation

Definition (Remainder). The error from approximating $f(x)$ by its tangent line around x^* is

$$R(x \mid x^*) := f(x) - L(x \mid x^*).$$

Thus

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + R(x \mid x^*).$$

Write

$$x = x^* + \Delta.$$

Then

$$R(x^* + \Delta \mid x^*) = f(x^* + \Delta) - f(x^*) - f'(x^*)\Delta.$$

Dividing by $\Delta \neq 0$ gives

$$\frac{R(x^* + \Delta \mid x^*)}{\Delta} = \frac{f(x^* + \Delta) - f(x^*)}{\Delta} - f'(x^*).$$

By differentiability,

$$\lim_{\Delta \rightarrow 0} \frac{f(x^* + \Delta) - f(x^*)}{\Delta} = f'(x^*).$$

Therefore,

$$\lim_{\Delta \rightarrow 0} \frac{R(x^* + \Delta \mid x^*)}{\Delta} = 0.$$

Equivalently,

$$R(x \mid x^*) = o(|x - x^*|) \quad \text{as } x \rightarrow x^*.$$

Intuition. The remainder goes to zero *faster* than the distance from x to x^* .

This is the precise sense in which the tangent line is the correct first-order approximation:

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \text{an error negligible relative to } |x - x^*|.$$

Can We Do Better? Higher-Order Polynomial Approximation

If information about higher derivatives is available, we can improve the linear approximation by using a polynomial.

Let

$$P(x \mid x^*, N) = \sum_{n=0}^N a_n (x - x^*)^n.$$

Explicitly,

$$P(x \mid x^*, N) = a_0 + a_1(x - x^*) + a_2(x - x^*)^2 + \cdots + a_N(x - x^*)^N.$$

We choose the coefficients so that P matches the value and the first N derivatives of f at x^* . For example,

$$P'(x) = a_1 + 2a_2(x - x^*) + 3a_3(x - x^*)^2 + \cdots + Na_N(x - x^*)^{N-1},$$

so

$$P'(x^*) = a_1.$$

Similarly,

$$P''(x^*) = 2!a_2,$$

$$P^{(3)}(x^*) = 3!a_3,$$

and, in general,

$$P^{(n)}(x^*) = n!a_n.$$

We impose

$$P^{(n)}(x^*) = f^{(n)}(x^*), \quad n = 0, \dots, N,$$

where

$$f^{(0)} := f.$$

Therefore,

$$n!a_n = f^{(n)}(x^*),$$

and hence

$$a_n = \frac{f^{(n)}(x^*)}{n!}.$$

Definition (Taylor polynomial). The *Taylor polynomial of order N* for f around x^* is

$$P(x \mid x^*, N) = \sum_{n=0}^N \frac{f^{(n)}(x^*)}{n!} (x - x^*)^n.$$

By construction,

$$P^{(n)}(x^*) = f^{(n)}(x^*), \quad n = 0, \dots, N.$$

Intuition. The Taylor polynomial does not merely pass through the point

$$(x^*, f(x^*)).$$

It matches progressively more local information:

$$\begin{aligned} N = 0 &: \text{ value,} \\ N = 1 &: \text{ value and slope,} \\ N = 2 &: \text{ value, slope, and curvature,} \\ &\vdots \end{aligned}$$

The higher the order, the more derivatives of f and P agree at the expansion point.

Example: Taylor Polynomials for the Exponential Function

Consider

$$f(x) = e^x$$

and approximate it around

$$x^* = 0.$$

Since

$$f^{(n)}(x) = e^x \quad \text{for every } n,$$

we have

$$f^{(n)}(0) = 1.$$

Hence

$$P(x \mid 0, N) = \sum_{n=0}^N \frac{x^n}{n!}.$$

The first few approximations are:

$$P(x \mid 0, 0) = 1,$$

$$P(x \mid 0, 1) = 1 + x,$$

$$P(x \mid 0, 2) = 1 + x + \frac{x^2}{2},$$

$$P(x \mid 0, 3) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6}.$$

More generally,

$$P(x \mid 0, N) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^N}{N!}.$$

Recall that

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Thus the Taylor polynomial is obtained by truncating this infinite series after finitely many terms.

Numerical illustration. At

$$x = -\frac{1}{2},$$

the true value is

$$e^{-1/2} \approx 0.60653.$$

The successive Taylor approximations are

$$P\left(-\frac{1}{2} \mid 0, 0\right) = 1,$$

$$P\left(-\frac{1}{2} \mid 0, 1\right) = 1 - \frac{1}{2} = \frac{1}{2},$$

$$P\left(-\frac{1}{2} \mid 0, 2\right) = 1 - \frac{1}{2} + \frac{1}{8} = \frac{5}{8} = 0.625,$$

and

$$P\left(-\frac{1}{2} \mid 0, 3\right) = 1 - \frac{1}{2} + \frac{1}{8} - \frac{1}{48} = \frac{29}{48} \approx 0.60417.$$

The approximation improves as additional local information is incorporated.

Taylor's Theorem with Lagrange Remainder

The Taylor polynomial gives the approximation. Taylor's theorem tells us exactly what form the approximation error takes.

Theorem 1 (Taylor's Theorem). *Let*

$$[a, b] \subseteq X \subseteq \mathbb{R},$$

and suppose that $f^{(N)}$ is continuous on $[a, b]$ and that $f^{(N+1)}$ exists on (a, b) .

For any

$$x, x^* \in [a, b],$$

there exists a point ξ lying between x and x^ such that*

$$f(x) = \sum_{n=0}^N \frac{f^{(n)}(x^*)}{n!} (x - x^*)^n + \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x^*)^{N+1}.$$

The first term is the Taylor polynomial:

$$P(x \mid x^*, N) = \sum_{n=0}^N \frac{f^{(n)}(x^*)}{n!} (x - x^*)^n.$$

The second term is the *Lagrange remainder*:

$$R(\xi \mid x^*, x, N) = \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x^*)^{N+1}.$$

Thus

$$f(x) = P(x \mid x^*, N) + R(\xi \mid x^*, x, N).$$

Remark. Taylor's theorem is nonconstructive with respect to ξ : it guarantees that some ξ between x^* and x exists, but it does not generally tell us which one.

For approximation purposes, however, we often do not need to know ξ exactly. It is enough to find an upper bound for

$$|f^{(N+1)}(\xi)|$$

on the relevant interval.

Bounding the Taylor Remainder

Suppose

$$|f^{(N+1)}(z)| \leq M$$

for every z between x^* and x .

Then Taylor's theorem gives

$$|R(\xi \mid x^*, x, N)| = \left| \frac{f^{(N+1)}(\xi)}{(N+1)!} (x - x^*)^{N+1} \right|.$$

Therefore

$$|R(\xi \mid x^*, x, N)| \leq \frac{M}{(N+1)!} |x - x^*|^{N+1}.$$

This provides an explicit bound on the approximation error.

Example: How Many Terms Do We Need?

Suppose we want to approximate

$$e^{1/2}$$

using the Taylor polynomial around

$$x^* = 0,$$

and we want the error to satisfy

$$|R| < 10^{-4}.$$

For $f(x) = e^x$,

$$f^{(N+1)}(z) = e^z.$$

Since

$$0 \leq z \leq \frac{1}{2},$$

we have

$$e^z \leq e^{1/2}.$$

Therefore,

$$|R| \leq \frac{e^{1/2}}{(N+1)!} \left(\frac{1}{2}\right)^{N+1}.$$

We seek N such that

$$\frac{e^{1/2}}{(N+1)!} \left(\frac{1}{2}\right)^{N+1} \leq 10^{-4}.$$

For $N = 5$,

$$\frac{e^{1/2}}{6!} \left(\frac{1}{2}\right)^6 \approx 3.58 \times 10^{-5} < 10^{-4}.$$

Thus

$$\boxed{N = 5}$$

is sufficient.

Taylor Approximation Is Local

Consider again

$$f(x) = e^x$$

and its first-order Taylor approximation around zero:

$$P(x \mid 0, 1) = 1 + x.$$

Taylor's theorem gives

$$e^x = 1 + x + \frac{e^\xi}{2}x^2$$

for some ξ between 0 and x .

Thus

$$R(\xi \mid 0, x, 1) = \frac{e^\xi}{2}x^2.$$

If x is close to zero, the quadratic factor

$$x^2$$

makes the error small.

For example, at

$$x = 0.1,$$

we have

$$0 \leq \xi \leq 0.1,$$

so

$$|R| \leq \frac{e^{0.1}}{2}(0.1)^2,$$

which is small.

By contrast, at

$$x = 10,$$

the same first-order approximation

$$1 + x$$

is extremely poor.

Intuition. A Taylor polynomial is a *local* approximation around x^* .

Its accuracy depends on both:

1. the order N of the polynomial;
2. the distance

$$|x - x^*|.$$

Increasing the order may improve the approximation, but moving far from the expansion point can make even a good local approximation inaccurate.

The Remainder as a Little- o Term

Taylor's theorem also gives a useful asymptotic interpretation.

Suppose

$$x_k \rightarrow x^*.$$

For each k , Taylor's theorem gives some ξ_k between x_k and x^* such that

$$R_k = \frac{f^{(N+1)}(\xi_k)}{(N+1)!} (x_k - x^*)^{N+1}.$$

Because ξ_k lies between x_k and x^* ,

$$|\xi_k - x^*| \leq |x_k - x^*|.$$

Since

$$x_k \rightarrow x^*,$$

the Squeeze Theorem implies

$$\xi_k \rightarrow x^*.$$

If $f^{(N+1)}$ is continuous at x^* , then

$$f^{(N+1)}(\xi_k) \rightarrow f^{(N+1)}(x^*).$$

Now consider

$$\frac{R_k}{|x_k - x^*|^N}.$$

We have

$$\left| \frac{R_k}{|x_k - x^*|^N} \right| = \frac{|f^{(N+1)}(\xi_k)|}{(N+1)!} |x_k - x^*|.$$

The first factor remains bounded near x^* , while

$$|x_k - x^*| \rightarrow 0.$$

Hence

$$\boxed{\frac{R_k}{|x_k - x^*|^N} \rightarrow 0.}$$

Therefore,

$$\boxed{R_k = o(|x_k - x^*|^N)}.$$

Corollary (Local asymptotic form of Taylor's theorem). Under the regularity assumptions above,

$$\boxed{f(x) = \sum_{n=0}^N \frac{f^{(n)}(x^*)}{n!} (x - x^*)^n + o(|x - x^*|^N)}$$

as

$$x \rightarrow x^*.$$

Intuition. The phrase

$$R(x) = o(|x - x^*|^N)$$

means that the approximation error becomes negligible relative to

$$|x - x^*|^N$$

as we approach the expansion point.

Thus Taylor's polynomial captures all terms up to order N in the local behavior of the function.

Taylor's Theorem in $\mathbb{R}^2$

Taylor Polynomial for a Function of Two Variables

Let

$$f : U \rightarrow \mathbb{R}, \quad U \subseteq \mathbb{R}^2,$$

and fix

$$(x^*, y^*) \in U.$$

Suppose the relevant partial derivatives of f exist and are continuous.

For

$$(x, y) \in U,$$

the Taylor polynomial of total degree N around (x^*, y^*) is

$$P_N(x, y) = \sum_{\substack{m, n \geq 0 \\ m+n \leq N}} \frac{1}{m!n!} \frac{\partial^{m+n} f}{\partial x^m \partial y^n}(x^*, y^*) (x - x^*)^m (y - y^*)^n.$$

The restriction

$$m + n \leq N$$

means that we retain every term whose *total degree* is at most N .

Taylor's Theorem in $\mathbb{R}^2$

Theorem 2 (Taylor's Theorem in two variables). *Let*

$$f : U \rightarrow \mathbb{R}, \quad U \subseteq \mathbb{R}^2$$

be open, and suppose that the line segment joining

$$(x^*, y^*) \quad \text{and} \quad (x, y)$$

is contained in U .

Suppose all partial derivatives of f up to order $N + 1$ are continuous on a neighborhood of this segment.

Then there exists a point

$$(w, z)$$

on the line segment joining (x^, y^*) and (x, y) such that*

$$f(x, y) = P_N(x, y) + R_{N+1}(x, y),$$

where

$$P_N(x, y) = \sum_{\substack{m, n \geq 0 \\ m+n \leq N}} \frac{1}{m!n!} \frac{\partial^{m+n} f}{\partial x^m \partial y^n}(x^*, y^*) (x - x^*)^m (y - y^*)^n,$$

and

$$R_{N+1}(x, y) = \sum_{\substack{m, n \geq 0 \\ m+n = N+1}} \frac{1}{m!n!} \frac{\partial^{N+1} f}{\partial x^m \partial y^n}(w, z) (x - x^*)^m (y - y^*)^n.$$

Remark. The structure is exactly analogous to the one-dimensional case:

$$\boxed{\text{function} = \text{polynomial approximation} + \text{remainder}.}$$

The difference is that, with several variables, terms of the same total order include both own-variable derivatives and cross-partials.

Example: A Second-Order Approximation in $\mathbb{R}^2$

Consider

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad f(x, y) = x^2 y.$$

We want the Taylor approximation of order

$$N = 2$$

around

$$(x^*, y^*) = (1, -2).$$

First,

$$f(1, -2) = -2.$$

The first derivatives are

$$f_x(x, y) = 2xy, \quad f_y(x, y) = x^2.$$

At the expansion point,

$$f_x(1, -2) = -4, \quad f_y(1, -2) = 1.$$

The second derivatives are

$$f_{xx}(x, y) = 2y,$$

$$f_{yy}(x, y) = 0,$$

and

$$f_{xy}(x, y) = f_{yx}(x, y) = 2x.$$

Thus,

$$f_{xx}(1, -2) = -4,$$

$$f_{yy}(1, -2) = 0,$$

and

$$f_{xy}(1, -2) = 2.$$

Therefore the second-order Taylor polynomial is

$$\begin{aligned} P_2(x, y) = & -2 - 4(x - 1) + (y + 2) \\ & + \frac{-4}{2!}(x - 1)^2 + 2(x - 1)(y + 2) + \frac{0}{2!}(y + 2)^2. \end{aligned}$$

Hence

$$\boxed{P_2(x, y) = -2 - 4(x - 1) + (y + 2) - 2(x - 1)^2 + 2(x - 1)(y + 2).}$$

The Remainder

The third derivatives are

$$\begin{aligned}f_{xxx} &= 0, & f_{yyy} &= 0, \\f_{xyy} &= 0,\end{aligned}$$

and

$$f_{xxy} = 2.$$

Therefore the only nonzero third-order term in the Taylor remainder is associated with

$$m = 2, \quad n = 1.$$

Hence

$$R_3(x, y) = \frac{f_{xxy}(w, z)}{2!1!}(x - 1)^2(y + 2).$$

Since

$$f_{xxy}(w, z) = 2$$

everywhere,

$$R_3(x, y) = \frac{2}{2}(x - 1)^2(y + 2).$$

Thus

$$\boxed{R_3(x, y) = (x - 1)^2(y + 2).}$$

In this example the third-order Taylor formula is exact:

$$\boxed{x^2y = P_2(x, y) + (x - 1)^2(y + 2).}$$

This occurs because $f(x, y) = x^2y$ is itself a polynomial of total degree three, so all derivatives of order four and higher vanish.